Mellin Energy, Cube Completion, and Quantified Descent

A Bounded Ram Calculus Study of the Alternate 11/12 Quasi-Riemann Manuscript

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Abstract

The assigned source is OpenAI's alternate 11/12 argument, `paper2.tex`, not the companion 7/8 manuscript. We map its entire proof architecture and read selected substantial proof intervals, while retaining the arithmetic reflection, conductor uniformity, and global mean-square claims as imported inputs. The Ram Calculus contribution is a source-specific normalization and error analysis. We prove that the displayed theta-weight transform is a unitary involution on multiplicative L^2, derive its first small-scale residue and a moment-cancellation extension, lower the row-kernel separation cost from $2m+4$ to $2m+3$ derivatives, and conditionally lower the two-Poisson transfer cost from $4m+12$ to $4m+9$. We give an explicit descent-depth bound, a weighted-logarithmic cube inverse, and finite Gram residual certificates. These statements neither prove the completed sextic-twist mean square nor certify a zero-free half-plane, the exclusion of exceptional zeros, or RH. Contribution assessment: C2/5, proved local analytic and bookkeeping refinements do not close the decisive arithmetic mean-square obligations.

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1 Assignment, Evidence, and Status

The applicability entry is M006, family 3, applicability **L3**. Applicability measures relevance, not validity or realized contribution. The latter is assessed separately as C2. The pinned public source has commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, date October 5, 2026, 3988 physical lines, and 169005 bytes. Its SHA256 is

`d9a8f15aa770cf883d0eabd2b775fad694ce20b44cba7928f5c0c9a6d8750d4d.`

It claims nonvanishing of every finite-order Hecke L-function over $K = \mathbb{Q}(\sqrt{-3})$ for $\Re s > 11/12$. Throughout this report that is a *source claim*, not a theorem certified here. Provenance, publication, unchanged bytes, and successful compilation do not establish mathematical validity. The author's literal reads cover lines 1–238, 426–2686, and 3290–3733: 2943 distinct lines. The omitted intervals are 239–425, 2687–3289, and 3734–3988. Reading a proof interval is itself different from independently verifying all of its imported results. The parent's authenticated `batch35-source-identity.json` records the assigned subtree unchanged between the pinned base and the captured October 8 target `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The M006 entry and local hash agree. This is inherited captured-tree provenance, not this author's fresh live-branch verification.

The entire structure was mapped by section and label inspection. In particular, the omitted arithmetic appendix contains reciprocity, cusp coefficients, automorphy, and the local Fourier calculation. Its conclusions are not silently inferred from its headings. Source labels below refer to the pinned file, not to similarly named results in another manuscript.

Source lines	Role	Audit treatment
1–205	Introduction, claims, notation	Literal read; claims distinguished from conclusions of this report.
206–659	Five-step roadmap	206–238 and 426–659 read; 239–425 not literally read.
660–772	Mean square to nonvanishing	Conditional extraction and Mellin implication checked below.
773–1253	Arithmetic identities; Poisson initialization	Read; normalizations and exclusions examined. Arithmetic identities are imported, not independently proved in all local cases.
1254–1631	Completion, cube inversion, transfer statement, descent	Read and analytic deductions reconstructed; principal target of the local work.
1632–2216	Theta reflection and completed mean square	Read; dependency and coefficient-independence requirements identified. No independent reflection or large-sieve proof audit.
2217–2686	Two Poisson transfers	Read; scale algebra, cancellation ordering, and derivative costs examined. Paired arithmetic identities remain inputs.
2687–3480	Arithmetic and theta appendix	Structure mapped; only 3290–3480 literally read. Earlier local and cusp calculations not inspected.
3481–3651	Weight separation	Full literal read; a strengthened analytic lemma proved here.

Source lines	Role	Audit treatment
3652–3988	Bibliography and end	3652–3733 read; remaining entries structurally located, not read.

Private proof evidence. Actual local proof text, not catalogue summaries, was read in three originals: RamCalBooks volume 2, chapter 1, lines 447–543 and 1214–1285 (Mellin inversion, energy, and truncation); chapter 2, lines 273–360 and 827–910 (physical Gram coordinates and perturbation); and TR1674 revised version 2, lines 1–380 (arithmetic energy, quarter-scale decomposition, finite Gram forms, and logical boundaries). These are 721 distinct lines across three files. Their hashes and exact intervals are in the read ledger. No whole-book, whole-corpus, or full private-source reading is claimed.

Separation from M005. The parent reports that M005 treats the companion `paper.tex` and already contains a finite endpoint/capacity improvement from $\Delta/4$ to $4\Delta/27$. That result is not used or credited as an M006 contribution. This author has not audited M005; the present analysis concerns the different cube recursion and weight transform in `paper2.tex`.

2 The Exact Source Family

Let $\mathcal{O} = \mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, and $N(z) = |z|^2$. Ideal indices prime to 3 are represented by their primary generators. Let S be a fixed finite set of prime ideals containing those above 2 and 3 and the conductor of a fixed finite-order character ν. A star denotes squarefree ideal indices prime to S. Row indices k are nonzero elements of $\mathcal{O}$, not squarefree ideals. Their unit multiplicity must not be deleted without accounting for it.

The source uses zero-extended sextic residue characters χ_n, $\alpha(n) = n/|n|$, normalized cubic Gauss sums $\gamma_2(n)$, and

$$a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n), \qquad |a_\xi(n)| \le 1.$$

Here ξ ranges over a *fixed finite* ray class character group. Its size and modulus are allowed in constants. For a compact interval $I = [u, v] \subset (0, \infty)$ and $W \in C_c^\infty(I)$, the source family is

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) = \frac{1}{XF} \sum_{F \le N(f) < 2F}^{*} \sum_{0 < N(k) \le \mathcal{H}} \left| \sum_{n}^{*} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(N(n)/X) \right|^2. \tag{2.1}$$

Set $\Sigma = XF$ and $q = \mathcal{E}/\Sigma$. The target is $q \ll D^\varepsilon$, *not* $\mathcal{E} \ll 1$. The underlying unnormalized squared sum has target $D^\varepsilon\Sigma^2$. The family is tested on common smooth weights; it is not an unrestricted all-coefficient operator-norm assertion.

Write $\Psi_k(n) = \xi(n)\chi_n(k)\chi_n(f)^4$, with zero extension at S. The completed sum is

$$T(X; k, f) = \sum_{n}^{*} \sum_{b} \frac{\overline{\alpha(n)}\gamma_2(n)\Psi_k(n)\overline{\alpha(b)}^3\Psi_k(b)^3}{\sqrt{N(n)}N(b)} V_*(N(n)N(b)^3/X), \quad V_*(x) = \sqrt{x}W(x). \tag{2.2}$$

The b sum is over all primary indices prime to S, not only squarefree ones. If $P(X; k, f)$ denotes the uncompleted column divided by $\sqrt{X}$, then complete multiplicativity gives

$$T(X; k, f) = \sum_{b} \frac{w_{k,f}(b)}{N(b)} P(X/N(b)^3; k, f), \quad w_{k,f}(b) = \overline{\alpha(b)}^3\Psi_k(b)^3. \tag{2.3}$$

Its local finiteness follows from $N(n)N(b)^3 \le vX$. In particular the $b = 1$ term is exactly P, without an extra power of X.

Three distinct norms. The column normalization $X^{-1/2}$, the row-auxiliary averaging factor F^{-1}, and the multiplicative measure dx/x have different roles. The private Mellin theorem identifies $\int |f(x)|^2 x^{2c-1}\,dx$ with the line $\Re s = c$ norm. For $c = 0$ the measure is dx/x; for $c = 1/2$ it is dx. In particular

$$\|V_*\|^2_{L^2(dx/x)} = \int_0^\infty |W(x)|^2\,dx. \tag{2.4}$$

There is no extra Jacobian after setting $x = e^t$ in the Mellin integral.

3 Theta Weight: Exact Energy and a Sharper Expansion

This section proves facts about the explicitly displayed scalar transform. It does not assert that the source's arithmetic sums transform into it. Let

$$a = \frac{(2\pi)^4}{27}, \qquad Q(z) = \frac{\Gamma(7/6+z)\Gamma(5/6+z)}{\Gamma(7/6-z)\Gamma(5/6-z)}, \qquad \mathcal{M}V(z) = \int_0^\infty V(x)x^z\frac{dx}{x}.$$

For $V \in C_c^\infty((0,\infty))$, define the source weight operator

$$(\mathscr{H}V)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \mathcal{M}V(-it)Q(it)(ax)^{-it}\,dt. \tag{3.1}$$

The integrand is absolutely integrable: Mellin decay is rapid and $|Q(it)| = 1$. Define H^s_M, for $s \ge 0$, by

$$\|V\|^2_{H^s_M} = \frac{1}{2\pi}\int_{\mathbb{R}} (1+t^2)^s |\mathcal{M}V(it)|^2\,dt.$$

This is the ordinary Sobolev space after $x = e^y$, with Fourier convention $\widehat{g}(t) = \int g(y)e^{-ity}\,dy$. Thus $\mathcal{M}V(it) = \widehat{g}(-t)$.

Theorem 3.1 (Source-weight isometry and involution)**.** *For every $s \ge 0$, $\mathscr{H}$ extends uniquely to a complex-linear unitary involution on H^s_M. On the imaginary Mellin line,*

$$\mathcal{M}(\mathscr{H}V)(it) = a^{-it}Q(it)\mathcal{M}V(-it). \tag{3.2}$$

If Π_T is the projection to $|t| \le T$, then $\Pi_T\mathscr{H} = \mathscr{H}\Pi_T$ and, for $T > 0$,

$$\|\mathscr{H}V - \mathscr{H}\Pi_T V\|_{L^2(dx/x)} = \|V - \Pi_T V\|_{L^2(dx/x)} \le T^{-s}\,\||t|^s\mathcal{M}V(it)\|_{L^2(dt/(2\pi))}. \tag{3.3}$$

Proof. Mellin inversion in logarithmic coordinates gives (3.2). Complex conjugation of the gamma function on positive-real-part arguments gives $|Q(it)| = 1$, and $Q(it)Q(-it) = 1$. The phase a^{-it} has modulus one. Reflection $t \mapsto -t$ preserves both dt and $(1+t^2)^s$. Plancherel therefore proves the norm equality on a dense set and the continuous extension. Applying (3.2) twice cancels both phases and gamma quotients. The symmetric band projection commutes with reflection and multiplication. Finally $|t|^{-s} \le T^{-s}$ on the omitted band gives (3.3). □

For $V = V_*$, the transformed weight has exactly the dx energy of W in (2.4). This provides a normalization test and a log-frequency truncation budget with constant one. It is not a bound on values at the discrete arithmetic sampling points. Point evaluation is not bounded on $L^2(dx/x)$,

and the completed mean square cannot be substituted into this identity without further arithmetic information.

The source proves small-x decay $x^{1/4}$ in `eq:theta-weight-decay`. The same kernel admits the following refinement.

Theorem 3.2 (First residue and moment cancellation)**.** *Let $I \Subset (0,\infty)$ be fixed and $V \in C_c^\infty(I)$. For every integer $j \geq 0$, every $0 < \beta < 5/6$, and every $A > 0$, there is a finite integer J, independent of I, such that*

$$|(x\partial_x)^j \mathscr{H} V(x)| \leq C_{I,A,\beta,j} \|V\|_{C^J(I)} \min(x^\beta, x^{-A}). \tag{3.4}$$

For $5/6 < \beta < 7/6$ and $0 < x \leq 1$ there is the more precise expansion

$$(x\partial_x)^j \mathscr{H} V(x) = \left(\frac{5}{6}\right)^j \frac{\Gamma(1/3)}{\Gamma(2)\Gamma(5/3)} \mathcal{M}V(5/6)(ax)^{5/6} + O_{I,\beta,j}(x^\beta \|V\|_{C^J(I)}). \tag{3.5}$$

Thus $\mathcal{M}V(5/6) = 0$ permits any small-x exponent below $7/6$.

Proof. On a fixed vertical line $\Re z = \sigma$, away from gamma poles, Stirling's formula gives a polynomial bound of order 4σ in $|\Im z|$ for $Q(z)$. Repeated integration by parts in $y = \log x$ gives arbitrarily large polynomial decay for $\mathcal{M}V(-z)$, uniformly in any fixed real strip, using finitely many ordinary derivatives of V on I. These estimates justify the rectangular contour shifts, with horizontal edges tending to zero, and differentiation by $(x\partial_x)^j$, which multiplies by $(-z)^j$.

The first numerator pole is $z = -5/6$; the next is $z = -7/6$. The reciprocal denominator gamma functions are entire. Shifting the line from zero to $-\beta$, with $0 < \beta < 5/6$, crosses no pole and gives x^β. Shifting to $A > 0$ crosses none and gives x^{-A}. For $5/6 < \beta < 7/6$, precisely the pole at $-5/6$ is crossed. Its residue is $\Gamma(1/3)\mathcal{M}V(5/6)(ax)^{5/6}/[\Gamma(2)\Gamma(5/3)]$. The differentiated residue has the factor $(5/6)^j$; the shifted integral is the remainder asserted. All support constants remain attached to I. □

This also identifies why the leading exponent is generally $5/6$, not an arbitrary positive power. If $V = \sqrt{x}W$, the cancellation condition is

$$\mathcal{M}V(5/6) = \int_0^\infty W(x) x^{1/3}\, dx = 0.$$

It can be imposed by subtracting a suitable multiple of a second smooth bump, but not by a nonzero nonnegative W. For $0 < \eta \leq 1$, the generic bound yields $\int_0^\eta |\mathscr{H} V|^2\, dx/x \leq C^2\eta^{2\beta}/(2\beta)$. This improves a low-scale quadrature tail; it does not remove a cusp term from the source's arithmetic transformation or improve the $11/12$ exponent.

4 Cube Inversion as an Operator, with Its Boundary

For any completely multiplicative w on ideals prime to S, with $w(1) = 1$ and $|w| \leq 1$, the finite identity underlying (2.3) has inverse

$$P(X) = \sum_d \frac{\mu(d)w(d)}{N(d)} T(X/N(d)^3). \tag{4.1}$$

Indeed the total cube index $b = dc$ has coefficient $w(b)\sum_{d|b}\mu(d)$, which is zero unless $b = 1$. This argument requires complete multiplicativity also at shared prime factors and at zeros; mere coprime multiplicativity would not suffice.

Proposition 4.1 (Damped logarithmic completion)**.** *For $a_0 > 0$, define*

$$\mathcal{L}_{a_0} = L^2((0,\infty), X^{-2a_0}\,dX/X), \qquad \mathscr{C}_w P(X) = \sum_b \frac{w(b)}{N(b)} P(X/N(b)^3).$$

The series converges in operator norm on $\mathcal{L}_{a_0}$. Its inverse is the operator in (4.1), also convergent in operator norm, and

$$\|\mathscr{C}_w\| \le \zeta_K(1+3a_0), \tag{4.2}$$

$$\left\|\mathscr{C}_w^{-1}\right\| \le \sum_b \frac{|\mu(b)|}{N(b)^{1+3a_0}} = \frac{\zeta_K(1+3a_0)}{\zeta_K(2+6a_0)}. \tag{4.3}$$

The sums on the right may include all ideals, giving upper bounds for the S-restricted operators. For $g(y) = e^{-a_0 y}P(e^y)$, the Fourier multiplier of $\mathscr{C}_w$ is the absolutely convergent Euler series

$$\sum_{(b,S)=1} \frac{w(b)}{N(b)^{1+3a_0+3it}} = \prod_{p\notin S}(1 - w(p)N(p)^{-1-3a_0-3it})^{-1}. \tag{4.4}$$

Proof. After weighting, the b term is $w(b)N(b)^{-1-3a_0}g(y - 3\log N(b))$. Translation is unitary on $L^2(dy)$. Absolute summability follows from convergence of the ideal zeta function for real part greater than one. The triangle inequality gives the operator bounds. Absolute convergence justifies multiplying the two series and grouping by $b = dc$; the same divisor identity proves that their product is the identity in either order. Taking the Fourier transform proves (4.4). The squarefree Euler product gives the displayed zeta ratio for the inverse majorant. □

This is a stable transform model, not an analytic continuation argument. An actual untruncated arithmetic P need not belong to $\mathcal{L}_{a_0}$ for every $a_0 > 0$; for example its elementary $O(X^{1/2})$ bound at infinity only supplies membership there for $a_0 > 1/2$, together with vanishing near zero. The abstract statement applies to members of the stated space, or compact-log-scale inputs. There is no uniform bound as $a_0 \downarrow 0$. For $w = 1$, even a finite Euler product has its maximal multiplier at $t = 0$; these maxima diverge as the prime set grows. Thus undamped completion on $L^2(dX/X)$ has no uniform bound obtainable from this argument. Damping cannot replace the short/long cube procedure at a fixed arithmetic scale.

4.1 Short and long cubes with the source normalizations

The imported completed estimate, `prop:R`, is

$$\sum_{0<N(k)\ll\mathcal{H}} |T(X;k,f)|^2 \ll_{I,\xi,S,\varepsilon,C_0} D^{\varepsilon}\,\|W\|^2_{C^J(I)}\left(\mathcal{H} + \frac{\mathcal{H}^2 N(f)}{X}\right), \tag{4.5}$$

for $1 \le \mathcal{H}, X, N(f) \le D^{C_0}$. It is not proved here. For the rest of this subsection assume $1 \le \mathcal{H}, X, F \le D^{C_0}$ and $\mathcal{H} \le \Sigma = XF$, as in the eventual positive-gap regime. Set $H_c^3 = \min(X, X^2/\mathcal{H}^2)$, $L_b = X/N(b)^3$. The averaging factor $1/F$, the number $O(F)$ of auxiliary indices, and $N(f) \asymp F$ give the short bound

$$\mathcal{E}_{\text{short}} \ll D^{\varepsilon}\,\|W\|^2_{C^J}\left(\mathcal{H} + \frac{\mathcal{H}^2 F H_c^3}{X}\right) \ll D^{\varepsilon}\Sigma\,\|W\|^2_{C^J}\,.$$

The reciprocal ideal sum costs a logarithm. If $H_c < 1$, the short sum is empty; the cutoff must not be artificially replaced by 1.

In the long part the coefficient is

$$\beta_0(b,f) = \overline{\alpha(b)}^3 \xi(b)^3 \mathbf{1}_{(b,f)=1} \sum_{\substack{d \mid b \\ N(d) > H_c}} \mu(d), \qquad |\beta_0(b,f)| \le \tau(b).$$

Since $\sum_{N(b)\le R} \tau(b)/N(b) \ll \log^2(2R)$, two weighted Cauchy–Schwarz factors yield at most a fourth power of a logarithm before absorption. For $L_b \le 1$, nonempty support forces $L_b \ge 1/v$ and finitely many columns, so $\mathcal{E}(\mathcal{H}, L_b, F) \ll_I \mathcal{H}/L_b \ll_I \Sigma$. This accounts for an endpoint sometimes lost by assuming $L_b \ge 1$. Consequently, *conditional on* (4.5),

$$\mathcal{E}(\mathcal{H}, X, F; W) \ll D^{\varepsilon} \left(\Sigma \, \|W\|_{C^J}^2 + \sup_{N(b) > H_c,\, L_b > 1} \mathcal{E}(\mathcal{H}, L_b, F; W) \right). \tag{4.6}$$

The supremum is empty if $\mathcal{H}^2 \le X$. One cannot discard $b \ne 1$ in T: completion is a signed complex sum and its terms can cancel. The private quarter-scale identity in TR1674 makes the analogous distinction between a contractive carried term and its uncontrolled cross/noise channels. That analogy is useful bookkeeping, not a transfer of Liouville cancellation to sextic twists.

5 A Lower-Derivative Row-Kernel Lemma

The source's even-order Laplacian proof in `lem:smooth-mean-square` requires $2m + 4$ derivatives. Coordinatewise integration by parts removes the parity restriction and gives a strict local improvement.

Theorem 5.1 (Odd-order separation)**.** *Fix compact intervals $I \subset \operatorname{int} I_* \Subset (0,\infty)$, an integer $m \ge 0$, a finite row set, nonnegative w_r, and finite sums*

$$S_{j,r}(U) = \sum_n a_{j,r}(n) U(x_{j,r,n}), \qquad j = 1, 2.$$

Assume for every $U \in C_c^\infty(I_*)$ that $\sum_r w_r |S_{j,r}(U)|^2 \le M_j \, \|U\|_{C^m(I_*)}^2$. For $|c_r| \le w_r$ and smooth K_r supported in I^2,

$$\left| \sum_r c_r \sum_{n,l} a_{1,r}(n) \overline{a_{2,r}(l)} K_r(x_{1,r,n}, x_{2,r,l}) \right| \le C_{I,I_*,m} \sqrt{M_1 M_2} \sup_r \|K_r\|_{C^{2m+3}(I^2)}. \tag{5.1}$$

The constant is independent of the number of rows, sample locations, and coefficients. The same assertion holds for countably many rows whenever the stated hypotheses and absolute limiting procedure apply.

Proof. Put $L_r(y,z) = K_r(e^y, e^z)$. Fourier inversion gives $K_r(x,y) = \int b_r(s,t) x^{is} y^{-it} \, ds \, dt$ with the usual $(2\pi)^{-2}$ factor in b_r. For any integer $q \ge 1$, choose a frequency coordinate whose absolute value is at least $(|s|+|t|)/2$. Integrate by parts q times in that coordinate. For small frequencies use the L^1 bound. Compact support eliminates boundary terms, and change to logarithmic coordinates costs only an I, q constant. Hence

$$\sup_r |b_r(s,t)| \le C_{I,q} (1 + |s| + |t|)^{-q} \sup_r \|K_r\|_{C^q}. \tag{5.2}$$

Unlike the source's Laplacian step, this holds for odd q.

Choose $V \in C_c^\infty(I_*)$, real, equal to 1 on I, and let $U_t(x) = V(x)x^{it}$. Multiplication by $V(x)V(y)$ does not change K_r. Then $\|U_t\|_{C^m} \le C(1+|t|)^m$. Take the *common pointwise* majorant (5.2) outside the row sum before integrating. Cauchy–Schwarz in r and the two mean-square hypotheses bound the desired expression by

$$C\sqrt{M_1 M_2}\sup_r \|K_r\|_{C^q} \int_{\mathbb{R}^2} \frac{(1+|s|)^m(1+|t|)^m}{(1+|s|+|t|)^q}\,ds\,dt.$$

For $q = 2m+3$ this integral is at most $4\int_0^\infty r(1+r)^{-3}\,dr = 2$. This proves the finite assertion and justifies absolute integral recombination. Finite row truncation followed by the indicated dominated limit gives the countable version. □

The common majorant is important: a separate L^1 or L^2 bound for each b_r does not automatically bound the integral of $\sup_r |b_r|$. At $q = 2m+2$, the radial majorant has logarithmic divergence. This proof does not assert that $2m+3$ is an optimal regularity threshold.

Corollary 5.2 (Conditional transfer derivative cost)**.** *Retain the source's arithmetic identities, index domains, fixed character families, Poisson transformations, and uniform kernel bounds in* `lem:first-transfer` *and* `lem:second-transfer`*. Replacing their uses of the source separation lemma by Theorem 5.1 yields*

$$\mathcal{A}(W) \ll D^{\varepsilon}\Sigma\,\|W\|^2_{C^{4m+9}(I)}\left(1+\sup\frac{\mathcal{E}(\mathcal{H}',X',F';\xi',U)}{\Sigma'}\right), \tag{5.3}$$

with exactly the source's child family, support $I' = [u/16, 4v]$, $\|U\|_{C^m(I')} \le 1$, and

$$\mathcal{H}' \le \frac{\mathcal{H}L}{\Sigma F}, \qquad \frac{\mathcal{H}'}{\Sigma'} \le \frac{\mathcal{H}}{\Sigma}, \qquad 1 \le \mathcal{H}', X', F', \quad \Sigma' = X'F' \le L. \tag{5.4}$$

Proof. The second separation now takes $2m+3$ derivatives of U, in place of $2m+4$. The first takes $2j+3$ of W with $j = 2m+3$, hence $4m+9$. The source's chain-rule kernel estimates hold at each integer order, so neither operation introduces another derivative. All other scale, divisor, positivity, and arithmetic steps are retained as hypotheses. Similarly the initialization cost $2J+4$ becomes $2J+3$ wherever that same row-kernel lemma is its sole reason for the derivative count. □

This is a conditional refinement of a specified proof component, not a new unconditional arithmetic transfer theorem.

6 Descent with Explicit Depth and Loss Budgets

Assume $D \ge 2$, $\kappa > 0$, $C_0 \ge 1$, and

$$\mathcal{H}, X, F \ge 1, \quad \Sigma = XF \le D^{C_0}, \quad \mathcal{H} \le \Sigma D^{-\kappa}. \tag{6.1}$$

If $\kappa > C_0$ these conditions are empty. For $\kappa = C_0$ they force the terminal case $\mathcal{H} = 1$. Below suppose $\kappa \le C_0$.

Proposition 6.1 (Full-contraction depth)**.** *Under the source cube reduction and transfer child inequalities, every nonterminal edge satisfies*

$$\mathcal{H}' < \mathcal{H}\left(\frac{\mathcal{H}}{\Sigma}\right)^2 \le \mathcal{H}D^{-2\kappa}. \tag{6.2}$$

The child retains (6.1). *Thus a branch needs at most*

$$N = \left\lceil \frac{C_0 - \kappa}{2\kappa} \right\rceil \tag{6.3}$$

transfer stages before its row bound is at most one (or the child set is empty). For a particular starting row range, replace N *by* $\lceil \log_D \mathcal{H}/(2\kappa) \rceil$. *One may stop earlier whenever* $\mathcal{H} \le F$ *or* $\mathcal{H}^2 \le X$.

Proof. Nonempty long-cube supremum implies $\mathcal{H}^2 > X$, whence $H_c^3 = X^2/\mathcal{H}^2$. Since $N(b) > H_c$ and $L = X/N(b)^3$, the first child inequality gives

$$\mathcal{H}' \le \frac{\mathcal{H}}{N(b)^3 F^2} < \frac{\mathcal{H}^3}{X^2 F^2}.$$

This remains valid when $H_c < 1$, including the possible index $b = 1$; in that regime the factor F^{-2} supplies the necessary decrease. The other child inequalities give $\mathcal{H}'/\Sigma' \le D^{-\kappa}$ and $\Sigma' \le L \le X \le \Sigma \le D^{C_0}$. The initial row exponent is at most $C_0 - \kappa$. After d edges it is at most $C_0 - \kappa - 2d\kappa$, proving the depth claim. A child would have to satisfy $\mathcal{H}' \ge 1$, so a strict bound below 1 actually makes the child family empty.

Elementary ideal and lattice counting gives $\mathcal{E} \le C_I \mathcal{H} X \, \|W\|_\infty^2$, hence $q \le C_I(\mathcal{H}/F) \, \|W\|_\infty^2$. This proves the early terminal criterion $\mathcal{H} \le F$. The other follows from the empty long-cube supremum in (4.6), conditionally on its completed input. □

The source uses levels spaced by κ and finally $\lceil C_0/\kappa \rceil$. The result above uses the full contraction already present in its inequality; it is a better depth certificate, not a stronger row-contraction theorem.

6.1 A non-asymptotic bookkeeping form

Normalize test functions by the required derivative norm. At an internal node v suppose the actual two estimates have been written as

$$q_v \le A_v D^{\delta_v}(1 + \sup_b r_{v,b}), \qquad r_{v,b} \le B_v D^{\delta_v}(1 + \sup_w q_w),$$

with $A_v, B_v \ge 1$ uniform over its children. Then, setting $K_v = 2A_v B_v$,

$$q_v \le K_v D^{2\delta_v}(1 + \sup_w q_w). \tag{6.4}$$

This formula exposes the otherwise hidden repeated constants. For a tree of depth at most N, terminal bound C_*, uniform $K_v \le K$, and $\delta_v = \varepsilon/(2N)$ when $N \ge 1$, induction yields

$$q_{\text{root}} \le (N + C_*) K^N D^\varepsilon. \tag{6.5}$$

Indeed unrolling a path gives at most N partial products and one terminal product, each bounded by $K^N D^\varepsilon$. For variable constants the same proof uses the maximum of the corresponding path products and requires every path sum $\sum_v 2\delta_v \le \varepsilon$.

There is a derivative budget as well. If Q is an upper bound for the cube lemma's derivative order at the chosen per-stage losses, take

$$J_0 = 1, \qquad J_{d+1} = \max\{Q, 4J_d + 9\}.$$

Putting $M = \max(1, Q)$ gives the explicit upper bound

$$J_d \le 4^d(M+3) - 3. \tag{6.6}$$

For the original $4m + 12$ cost the comparable bound is $4^d(M+4) - 4$. For example $M = 1$, $d = 2$ gives 61 instead of 76. At depth d, supports lie inside $[u/16^d, 4^d v]$; constants must be taken on those actual intervals, not on the original I alone. Since N is fixed once κ, C_0 are fixed, this is a legitimate D-uniform conditional conclusion. It is not uniform as $\kappa \to 0$. No growing-depth scheme follows merely from writing D^ε at each step. Actual numerical K and Q are not provided by this report.

7 Poisson Energy and the Arithmetic Obligations

The source's lattice convention is

$$e(z) = \exp(4\pi i \operatorname{Im} z/\sqrt{3}), \quad d\mu(z) = \frac{2}{\sqrt{3}}\,dx\,dy, \quad \widehat{\Phi}(|w|^2) = \int_{\mathbb{C}} \Phi(|z|^2) e(-zw)\,d\mu(z).$$

The lattice $\mathcal{O}$ has covolume one for $d\mu$; this pairing makes it self-dual. For a primitive character χ of ideal modulus $\mathfrak{m}$ and an exclusion ideal $\mathfrak{r}$, the source's Poisson formula has factor

$$\frac{\mathcal{H}\gamma(\chi)}{\sqrt{N(\mathfrak{m})}} \sum_{d \mid \operatorname{rad}\mathfrak{r}} \frac{\mu(d)\chi(d)}{N(d)} \sum_h \overline{\chi(h)} \widehat{\Phi}\left(\frac{\mathcal{H}N(h)}{N(d)N(\mathfrak{m})}\right).$$

Here $k = d\ell$ changes the row scale to $\mathcal{H}/N(d)$, not to $\mathcal{H}/N(d)^2$. Nonprincipal primitive characters have zero zero-frequency coefficient; the principal primitive modulus-one case has the Euler factor $\prod_{p \mid \mathfrak{r}}(1 - N(p)^{-1})$. These are normalization checks, not an independent verification of every paired Gauss-sum identity used later.

7.1 The two transfers

The first source transfer enlarges a *nonnegative* intermediate mean square by $\Sigma/(LF) \ge 1$. Signed off-diagonal sums cannot be enlarged that way before they have been put into the requisite form. In the second transfer, the reindexing $r = tg/e$, $f' = Cew$, $k' = deh$ gives

$$X_0' = \frac{L}{N(r)N(f')}, \quad w_{C,d}\frac{Y_{C,d}N(g)}{N(e)\ell} = \frac{c_I \Sigma N(r)}{L^2}.$$

The exact preimage cancellation at $p \mid f'$ is

$$(1 + \mathbf{1}_{p \mid k'}) - \mathbf{1}_{p \mid k'} - \mathbf{1}_{p \nmid k'} = \mathbf{1}_{p \mid k'}. \tag{7.1}$$

Taking absolute values before this regrouping would lose $f' \mid k'$. That divisibility is subsequently used to prove $X'/N(j) \ge 1$ for every exclusion divisor $j \mid r$. This is a substantive dependency, not an optional simplification of the proof.

At dyadic $R \le N(r) < 2R$, $F' \le N(f') < 2F'$, the source has

$$X' = L/(RF'), \quad \Sigma' = L/R, \quad \mathcal{H}' = \mathcal{H}L/(\Sigma F R^2).$$

Thus $\mathcal{H}'/\Sigma' = \mathcal{H}/(\Sigma F R)$ and $(\Sigma R/L^2)R(\Sigma')^2 = \Sigma$. These identities explain both the child gap and why one must keep F in the transfer normalization. The zero-frequency estimate uses $Y_{C,d} > 0$, not $Y_{C,d} \ge 1$.

7.2 What remains imported

1. **Arithmetic conversion.** The identities converting paired sextic Gauss sums into a_ξ and back, including supplementary ray factors, are not established by positivity or Fourier unitarity. The relevant unread appendix portions remain explicit inputs.

2. **Reflection with uniform indexing.** In source `lem:reflection-uniformity`, the cusp coefficient, additive character, and fixed conductor component must be constant on a fixed finite partition of the varying squarefree row k_0. If they secretly vary with its full value, the subsequent all-coefficient quadratic large sieve cannot be applied with a row-independent coefficient sequence.

3. **Quadratic Hecke family.** Goldmakher–Louvel's theorem is for an appropriate primitive quadratic Hecke family, with squarefree indices and fixed reciprocity data. It is not a theorem for arbitrary sextic characters. The source's construction of that family and its exclusions need an independent local check.

4. **Bad primes and repeated rows.** The source's local updates for exponent $j_p = 4$ have three branches, including $q^{-1/2}$ factors. Their combined amplitude–length invariant is $a_\iota^2 Y_\iota$, not the dual length alone. For $k = u_0 s v^2$, the auxiliary twist becomes $f v^2$, and the final summation is $\sum_v N(v)^{-2} = \zeta_K(2)$. Omitting this change or using $N(v)^{-1}$ would break the stated bound.

5. **Uniformity in all varying scales.** Source constants may depend on $S, \nu, I, C_0, \varepsilon$, the fixed ray group, and a fixed positive gap. They may not acquire uncontrolled dependence on k, f, X, or the cube index. No effective numerical values for these constants are certified here.

We found no contradiction in the limited scale algebra checked above. That is not a finding that the full proof is correct. In particular the unitary scalar transform of Section 3 leaves all five arithmetic obligations intact.

8 Finite Ram Gram Certificates and Their Limits

For fixed finite row and column ranges from (2.1), define

$$A_{(f,k),n} = \frac{a_\xi(n)\chi_n(k)\chi_n(f)^4}{\Sigma}, \qquad z_n = W(N(n)/X).$$

With Euclidean coordinate norms one has exactly

$$q = \|Az\|_2^2 = z^* A^* A z, \qquad A^*A \succeq 0. \tag{8.1}$$

This is the arithmetic family version of the finite Gram formulation read in TR1674. Its exactness is linear algebra; it supplies no cancellation beyond that present in the entries. A Parseval change of row coordinates preserves q; a non-tight frame requires its upper and lower frame bounds.

Proposition 8.1 (Finite residual enclosure)**.** *Let $\widetilde{y}$ be a computed or approximated version of Az, and suppose a* proved *bound $\|Az - \widetilde{y}\|_2 \le \delta$ is available. Then*

$$\max(0, \|\widetilde{y}\|_2 - \delta)^2 \le q \le (\|\widetilde{y}\|_2 + \delta)^2. \tag{8.2}$$

If $\widetilde{y} = \widetilde{A}\widetilde{z}$, $\left\|A - \widetilde{A}\right\|_2 \le \eta$, and $\|z - \widetilde{z}\|_2 \le e$, it suffices to take $\delta = \eta\,\|\widetilde{z}\|_2 + (\left\|\widetilde{A}\right\|_2 + \eta)e$.

Proof. The triangle and reverse triangle inequalities give the enclosure. Expand $Az - \widetilde{A}\widetilde{z}$ as $(A - \widetilde{A})\widetilde{z} + A(z - \widetilde{z})$ and bound $\|A\| \le \left\|\widetilde{A}\right\| + \eta$. □

Entrywise verified errors imply $\eta \le (\sum_{r,n} \eta_{rn}^2)^{1/2}$. Different numerical approximations alone do not provide such an enclosure. The finite positivity test also does not supply uniform estimates for all scales or all smooth tests. An all-ones model with $O(\mathcal{H}F)$ rows and $O(X)$ columns has the same elementary normalization and $q \asymp \mathcal{H}/F$; mere coefficient boundedness cannot improve it to $D^{o(1)}$ throughout the positive-gap regime. For $F = 1$, $\mathcal{H} = X^{1-\kappa}$, $0 < \kappa < 1$, the obstruction is a positive power of X. This is a counterexample to a proposed *generic analytic substitute*, not a counterexample to the actual arithmetic family.

9 Conditional Extraction and the 11/12 Boundary

The original mean-square proposition states, for every fixed $0 < \vartheta \le 1/10$, every $\varepsilon > 0$, and all smooth W on a fixed compact interval, a bound of the form

$$\sum_{0<N(u)\le D^{1+\vartheta}} |A_u(D)|^2 \ll \|W\|_{C^J}^2 D^{2+\vartheta+\varepsilon}, \quad A_u(D) = \sum_{(n,S)=1} \mu(n)\nu(n)\chi_n(u)W(N(n)/D). \tag{9.1}$$

This remains an unverified arithmetic input to the following deduction. Put $Y = D^{(1+\vartheta)/6}$ and take primes $Y/2 < N(p) \le Y$, outside S. Their sixth powers are distinct rows within the original range and $\chi_n(p^6) = \mathbf{1}_{p\mid n}$. Hence $|A_1 - A_{p^6}| \ll_W D/Y$. The prime ideal theorem for the fixed field, also an imported classical input, supplies $J_p \asymp Y/\log Y$ primes. Averaging and (9.1) give

$$|A_1(D)|^2 \ll D^{11/6+5\vartheta/6+\varepsilon} + D^{5/3-\vartheta/3}. \tag{9.2}$$

For any desired exponent loss choose a fixed sufficiently small $\vartheta > 0$, then a sufficiently small mean-square loss. This yields $A_1(D) \ll D^{11/12+\varepsilon}$ conditionally. One does not take a D-dependent $\vartheta \to 0$ with the same constants. The local derivative and depth improvements here do not change either exponent in (9.2).

For completeness, assume that last bound for every compactly supported smooth test. The integral

$$\int_0^\infty A_1(D) D^{-s} \frac{dD}{D}$$

then defines a holomorphic function for $\Re s > 11/12$: on each compact subset choose the saving loss below its distance to the boundary, and at zero the integrand vanishes by support. For $\Re s > 1$, absolute convergence and $x = N(n)/D$ give $\widehat{W}(s)/L_K^S(s,\nu)$. At a hypothetical zero ρ in the half-plane, choose $W(x) = x^{-\rho}\phi(x)$ with $0 \ne \phi \in C_c^\infty((1,2))$, $\phi \ge 0$. Then $\widehat{W}(\rho) = \int \phi(x)\,dx/x > 0$. Meromorphic continuation and the identity theorem applied to $L_K^S \mathcal{M}_W = \widehat{W}$ exclude that zero. Possible poles are handled meromorphically, not by pretending L_K^S is holomorphic everywhere. Omitted Euler factors have no zeros in this positive-real-part half-plane.

This explains precisely where a zero-free conclusion would enter if the arithmetic input were established. It is not established by a finite matrix test, by the logarithmic completion operator, or by the theta-weight isometry. Dirichlet base change and exceptional-zero consequences therefore remain source-reported consequences, not deliverables of M006.

10 Public Literature Refresh and Research Priorities

Public-title-only searches and primary-page reads were made on October 8, 2026. Confidential passages and private titles were not used in searches. The live public README identifies the October 5 manuscript and its `paper2.pdf`; this verifies public identification, not proof validity or independent endorsement. The report's substantive analysis stays tied to the pinned local bytes.

Goldmakher and Louvel [2] formulate the quadratic large sieve with squarefree ideal indices and an explicitly defined Hecke family; the primary PDF introduction, definition, and theorem were read. Its full proof was not audited. Dunn and Radziwiłł [3] provide the relevant cubic Gauss-sum/Voronoi context. Their primary abstract explicitly states GRH dependence of the Patterson-bias asymptotic and dispersion estimate. This does not by itself make every theta identity in that paper conditional, nor does citing it establish the new source's required reflection with uniform varying twists.

Gao and Zhao [4] study a quadratic Möbius-twist mean square over positive odd squarefree integer rows; their abstract and bibliographic record were read, not the theorem proof. It is not a substitute for the sextic Eisenstein family (9.1). Mossinghoff, Trudgian, and Yang [5] give explicit height-dependent zero-free regions; the primary abstract was read. Those regions illustrate the difference between a shrinking width at large height and a fixed-width half-plane. We make no claim that this bounded search exhausts the literature or determines the best available constant on the refresh date.

10.1 Pros, costs, and next proofs

The exact energy identity catches measure and scale mistakes, and the residue expansion gives a concrete low-scale tail budget. Odd-order separation saves three derivatives per transfer stage. The full-contraction depth reduces repeated derivative growth and makes the dependence on a fixed gap explicit. Finite Gram certificates separate arithmetic entry errors, weight approximation, and numerical output error.

The costs are equally concrete. L^2 control alone does not bound sampled arithmetic sums; moment cancellation may require signed weights; damping changes the scale norm; the derivative order still grows exponentially with descent depth; and support constants may grow with that depth. None of these devices creates cancellation in a sextic-twist family.

The next highest-value verification is an independent reconstruction of the unread cusp/local-transform calculation, including the fixed-ray uniformity statement. It should be followed by a complete check of the quadratic Hecke-family hypotheses and all exponent-0 and exponent-4 local factors. A separate finite-arithmetic program could test those identities in small Eisenstein residue fields with exact algebraic coefficients. Such tests would be diagnostics, not a uniform proof. Other useful work is to make support-dependent constants effective, test whether more economical regularity norms survive both transfers, and quantify a true arithmetic improvement in (9.1). These are future tasks, not accomplishments credited in the present assessment.

11 Verification and Contribution Assessment

The accompanying bounded checks exercise cube-convolution inversion, child-scale identities, the preimage cancellation (7.1), the two derivative recurrences, finite Gram/residual inequalities, and the modulus/involution identity of the gamma multiplier. Exact rational checks are separated from floating-point diagnostics. They do not test the full Eisenstein Gauss-sum system, prove theta automorphy, verify large-sieve hypotheses, or certify an infinite-scale mean square.

The source and three private originals are hashed before their scientific reads and compared again at finalization. Equality of endpoints is not continuous monitoring. Native opening and compilation are attempted first; an existing MiKTeX export helper supplies the on-disk PDF if needed. The final PDF is rendered and every page is visually inspected. The audit files record compilation, tests, scope, assessment, and a frozen manifest. No shared queue, progress file, catalogue, helper, or other manuscript is edited by this author.

The realized contribution is **C2/5**, independent of applicability **L3**. Theorems 3.1, 3.2, and 5.1, Proposition 4.1, and the conditional Corollary 5.2 and Proposition 6.1 are useful source-relative analytic refinements. Classical Plancherel, contour shifting, Möbius inversion, Cauchy–Schwarz, and finite linear algebra are not claimed as inventions. No historical priority or substantial arithmetic advance is asserted. The important unresolved boundary is the completed and transferred arithmetic mean square, not the syntax of its energy representation. This report does not certify QRH or RH.

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